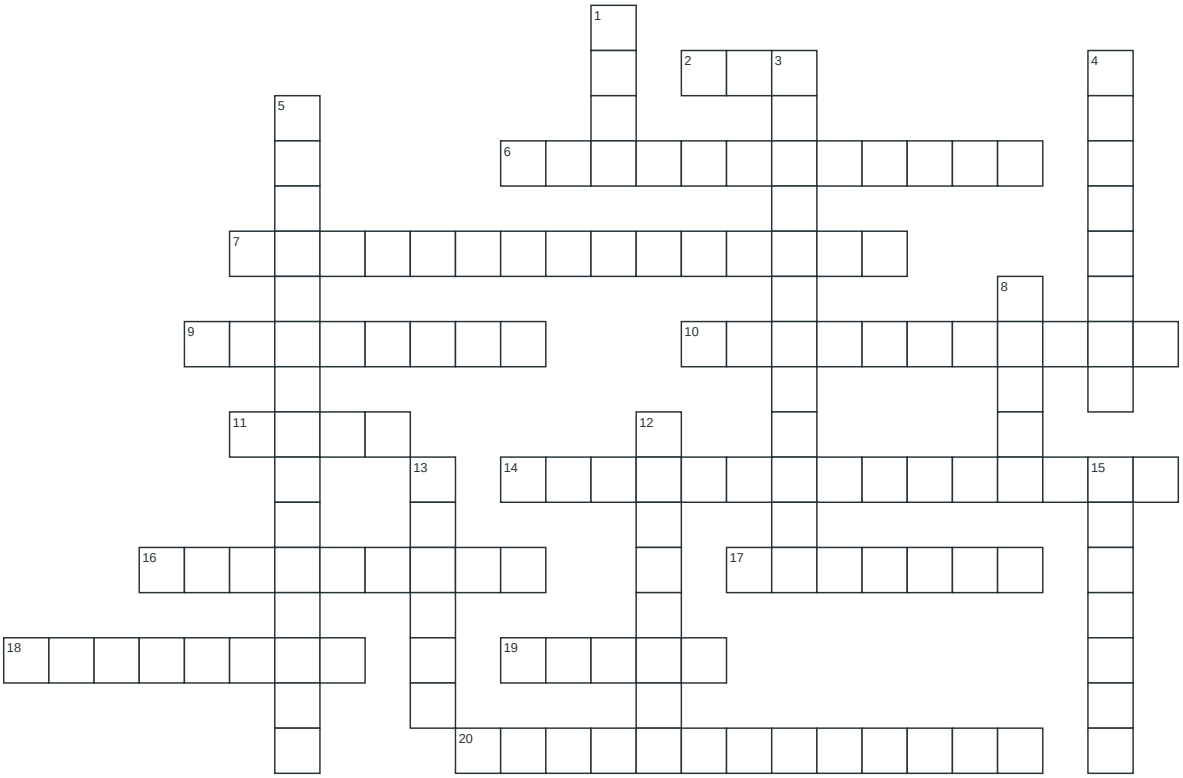
# Forces and Newton's Laws Crossword

Hard | 20 clues | Student Worksheet

Name: _____ Date: _____

Directions: Use the clues to complete the grid. Omit spaces, punctuation, and hyphens.



## ACROSS

**2.** Relationship that expresses Newton's second law for constant mass
**6.** The directed change from initial to final position
**7.** Equal and opposite interaction forces acting on different objects
**9.** The total length of a traveled path
**10.** Force that is a surface's push perpendicular to that surface
**11.** State in which position does not change relative to a chosen reference point.
**14.** Energy associated with position in a gravitational field
**16.** Orientation of motion, needed to describe a velocity vector.
**17.** Resistance to changes in motion
**18.** Opposes the relative sliding of contacting surfaces

## ACROSS (continued)

**19.** Quantity that is a scalar
**20.** Energy of motion; for a fixed mass it varies with the square of speed.

## DOWN

**1.** Measure of the amount of matter in an object
**3.** Change in velocity per unit time.
**4.** The vector sum of all forces acting on an object
**5.** SI-derived unit commonly used to measure speed or velocity.
**8.** Definition of the SI unit of energy and work
**12.** Exerts a force that changes the passenger's motion
**13.** The gravitational force acting on an object
**15.** In the ideal near-Earth projectile model, which force acts after launch when air resistance is neglected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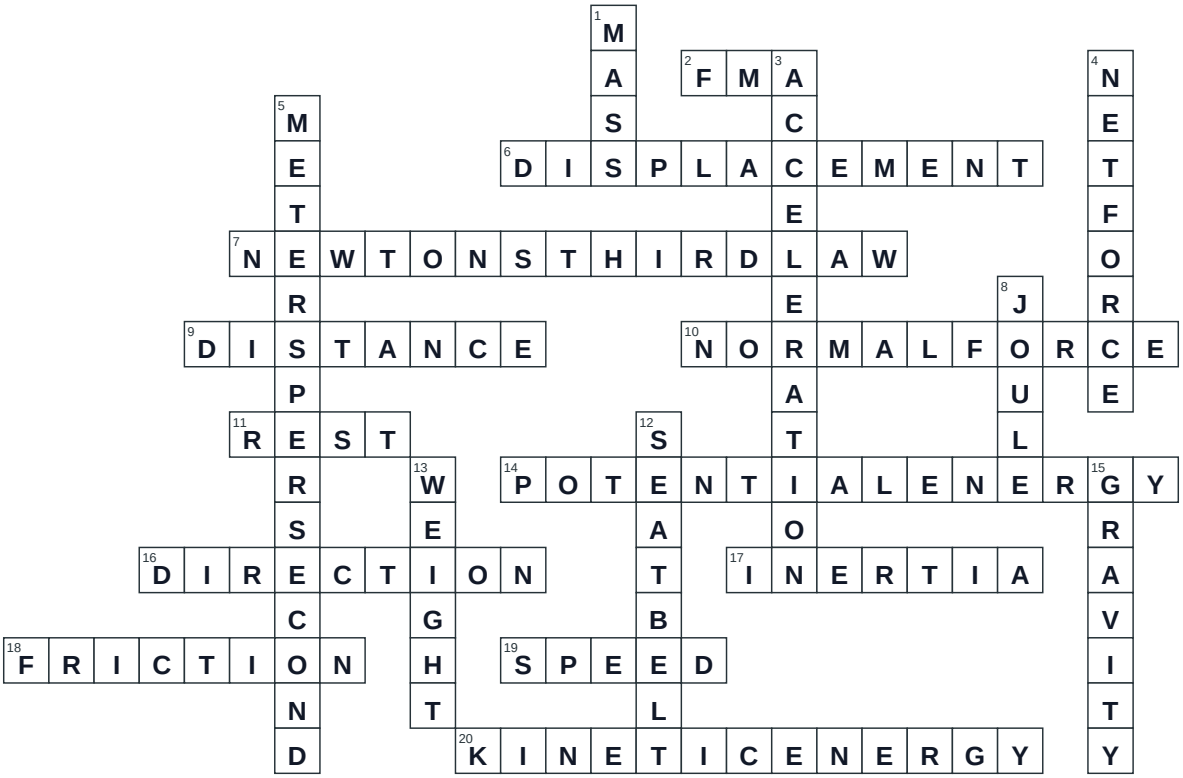
# Forces and Newton's Laws Crossword

Hard | 20 clues | Teacher Answer Key

### TEACHER ANSWER KEY

Completed grid and numbered answers. Spaces and punctuation are omitted in the grid.




## ACROSS

**2.** F = ma
**6.** Displacement
**7.** Newton's Third Law
**9.** Distance
**10.** Normal Force
**11.** Rest
**14.** Potential Energy
**16.** Direction
**17.** Inertia
**18.** Friction
**19.** Speed

## ACROSS (continued)

**20.** Kinetic energy

## DOWN

**1.** Mass
**3.** Acceleration
**4.** Net Force
**5.** Meters per second
**8.** Joule
**12.** Seat Belt
**13.** Weight
**15.** Gravity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